

Xiuyin Li



Seeking research-assistant roles | working toward an AI PhD

● Interests: Reliable & self-improving LLM agents: feedback-based post-training, verifiable reward modeling, synthetic data & environment construction, causal & uncertainty-aware evaluation

📄 IELTS: 7.0 (W: 7.0, S: 6.5) | CET-4: 591 | CET-6: 552

EDUCATION



The University of Hong Kong Master of Data Science **3.51/4.30** Sep 2025 – Dec 2026

Deep Learning (A+), NLP and Text Analytics (A+), Machine Learning (B+), Programming (A), Data Mining (A-), Statistical Inference (A-), Advanced Statistical Modelling (A-), Time Series Forecasting (B+)

Shandong University B.S. in Statistics **86.52** Sep 2021 – Jun 2025

Machine Learning (95), Python Programming (95), Statistical Computation (99), Bayesian Statistics (96), Discrete Mathematics (94), Mathematical Statistics (92), Multivariate Statistical Analysis (91), Mathematical Modeling (90), Advanced Algebra (86), Real Function Theory (86), Mathematical Analysis (86), Stochastic Processes (82)

B.E. in Computer Science (minor) **88.86** Sep 2022 – Jun 2025

Data Structures (87), Software Engineering (92), Operating System (91), Object-Oriented Technology (90)

RESEARCH EXPERIENCE

B.S. Thesis: EM and MCEM for Parameter Estimation in Regression Models May 2025

- Derived and implemented EM/MCEM with multicycle-ECM for Student's t-regression and linear mixed models in R, including REML corrections that remove the small-sample downward bias in variance components.
- Validated every estimator to machine precision against lme4, nlme, and optim over 500-replication simulations, then applied the methods to robust facial-age regression and a 60k-observation hierarchical reaction-time model.

SELECTED PROJECTS

Meeting Agent: Multimodal RAG with Long-Term Memory Project Lead · 2026

- Architected the core RAG retrieval funnel—scope routing (corpus → meeting → file), hybrid dense+sparse recall (Chroma + BM25/FTS5), Reciprocal Rank Fusion with per-file fair allocation, anchor-based eviction for session continuity, and Cohere/BGE reranking—yielding citation-grounded, traceable answers.
- Built a content-aware ingestion cascade with speaker-diarized ASR and a TTL/importance-decayed long-term memory with a knowledge graph of entities and relations; implemented with FastAPI and LangChain LCEL.

Zero-Shot CNNs from Textual Descriptions — Reproduction & Extension Project Lead · 2026

- Reproduced Ba et al. (ICCV 2015) in PyTorch—predicting convolutional and fully-connected classifier weights directly from Wikipedia text to recognize unseen classes on CUB-200-2011 and Oxford Flowers (0.71 unseen ROC-AUC)—then extended it with auxiliary losses, modern text encoders (SBERT/CLIP), and backbones (DenseNet-121, ResNet-50), reaching 0.89 unseen ROC-AUC with a CLIP text encoder.

CNN, BiLSTM & Transformer for Text Classification Personal · 2025

- Derived all three architectures from first principles—the recurrent family (S-RNN → LSTM → Peephole → BiLSTM) with gating and BPTT derivations, CNN inductive biases, and multi-head self-attention—then benchmarked them on four datasets (IMDB, AG News, BBC News, SST-5), tying the measured $\mathcal{O}(n^2)$ throughput drop to a complexity analysis and showing architecture-task fit outweighs capacity.

SELECTED AWARDS

MCM —Honorable Mention, *Momentum in Tennis* Feb 2024

CUMCM —First Prize (Shandong Province), *Vegetable Pricing and Replenishment* Sep 2023

SKILLS

Machine Learning & AI: PyTorch, NumPy, Pandas, SciPy; classical machine learning (scikit-learn); LLM fine-tuning (Hugging Face Transformers, PEFT/LoRA) and inference (vLLM); retrieval-augmented generation (LangChain, Chroma); vision-language models (CLIP, Qwen-VL); statistical modeling

Programming Languages: Python, R, SQL (*Proficient*); C++, Java (*Working Knowledge*); JavaScript (*Familiar*)

Tools & Systems: Linux (CLI), Git, CI/CD, Docker, conda, uv, W&B, pytest, L^AT_EX